

Implementation of Network Commands

Command: **Ipconfig**

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26100.7171]
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C:\Users\Exam25>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : MC.network
    Link-local IPv6 Address . . . . . : fe80::c78b:2c5f:2993:66b6%4
    IPv4 Address. . . . . : 10.10.14.164
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 10.10.1.1

C:\Users\Exam25>
```

- It shows the IP address assigned to my computer.
- IPv4 address identifies my device on the network.
- Default Gateway is the router through which my system connects to the internet.

Command: **nslookup google.com**

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26100.7171]
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C:\Users\Exam25>nslookup google.com
Server: UnKnown
Address: 10.10.1.6

Non-authoritative answer:
Name:   google.com
Addresses: 2404:6808:4802:81c::208e
          142.250.182.238

C:\Users\Exam25>
```

- This command checks how a domain name is converted into an IP address.
- It shows the DNS server used and the IP address of google.com.
- DNS helps browsers find websites on the internet

Command: **tracert google.com**

```
C:\Users\Exam25>tracert google.com

Tracing route to google.com [142.250.182.238]
over a maximum of 30 hops:

  0  14 ms  18 ms  4 ms  10.10.1.1
  1  45 ms  19 ms  19 ms  10.133.1.1
  2  23 ms  16 ms  16 ms  172.31.113.200
  3  29 ms  54 ms  30 ms  172.31.237.156
  4  39 ms  48 ms  65 ms  10.102.1.2
  5   3 ms   3 ms   3 ms  10.103.1.11
  6  24 ms   8 ms   3 ms  192.168.1.1
  7  10 ms  10 ms  26 ms  10.119.96.253
  8  *      *      *      Request timed out.
  9  9 ms   10 ms  11 ms  10.119.234.162
 10  48 ms  16 ms  19 ms  72.14.194.160
 11  39 ms  18 ms  27 ms  142.251.77.27
 12  10 ms  15 ms  12 ms  142.251.52.201
 13  62 ms  42 ms  20 ms  bom07s29-in-f14.1e100.net [142.250.182.238]

Trace complete.

C:\Users\Exam25>
```

- Tracert shows the path taken by data packets from my computer to the destination.
- Each hop represents a router in between.
- It helps identify where network delays or failures occur.

Command: **ping google.com**

```
C:\Users\Exam25>ping google.com

Pinging google.com [142.250.182.238] with 32 bytes of data:
Reply from 142.250.182.238: bytes=32 time=12ms TTL=110
Reply from 142.250.182.238: bytes=32 time=10ms TTL=110
Reply from 142.250.182.238: bytes=32 time=19ms TTL=110
Reply from 142.250.182.238: bytes=32 time=20ms TTL=110

Ping statistics for 142.250.182.238:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 20ms, Average = 15ms

C:\Users\Exam25>
```

- Ping checks whether my system can reach another computer or server.
- It measures the response time of data packets.
- Successful replies mean the network connection is working.